1.20 Inequalities

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Set Builder Notation

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Recall that a **set** is a **collection of values** such that:
a) The **order of values does not matter**.
b) There are **no duplicates**.

Recap from GCSE:
- We use curly braces to list the values in a set, e.g. $A = \{1, 4, 6, 7\}$
- If A and B are sets then $A \cap B$ is the **intersection** of A and B, giving a set which has the elements in A **and** B.
- $A \cup B$ is the **union** of A and B, giving a set which has the elements in A **or** in B.
- $\varnothing$ is the empty set, i.e. the set with nothing in it.
- Sets can also be infinitely large. $\mathbb{N}$ is the set of natural numbers (all positive integers), $\mathbb{Z}$ is the set of all integers (including negative numbers and 0) and $\mathbb{R}$ is the set of all real numbers (including all possible decimals).
- We write $x \in A$ to mean "x is a member of the set A". So $x \in \mathbb{R}$ would mean "x is a real number".
-

$$\{1, 2, 3\} \cap \{3, 4, 5\} =$$
$$\{1, 2, 3\} \cup \{3, 4, 5\} =$$
$$\{1, 2\} \cap \{3, 4\} =$$

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Set Builder Notation

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It is possible to construct sets without having to explicitly list its values. We use:

The | or : means "such that".

$$\{expr \mid condition\}$$
or $$\{expr : condition\}$$

Can you guess what sets the following give?

$$\{2x : x \in \mathbb{Z}\} = _____$$
$$\{2^x : x \in \mathbb{N}\} = _____$$
$$\{xy : x, y \text{ are prime}\} = _____$$

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Set Builder Notation

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Can you use set builder notation to specify the following sets?

All odd numbers.

All (real) numbers greater than 5.

All (real) numbers less than 5 **or** greater than 7.

All (real) numbers between 5 and 7 inclusive.

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Recap of linear inequalities

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Inequality	Solution Set
$2x + 1 > 5$	
$3(x - 5) \geq 5 - 2(x - 8)$	
$-x \geq 2$	

Note: Multiplying or both sides of an inequality by a negative number reverses the direction.

Combining Inequalities:

If $x < 3$ and $2 \leq x < 4$, what is the combined solution set?

2 3 4

If both inequalities have to be satisfied, we have to be on both lines. Place your finger vertically and scan across.

1.20 Inequalities

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RECAP :: Solving Quadratic Inequalities

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Solve $x^2 + 2x - 15 > 0$

$(x + 5)(x - 3) > 0$

Step 1: Get 0 on one side

Step 2: Factorise

Step 3: Sketch and reason

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Solving Quadratic Inequalities

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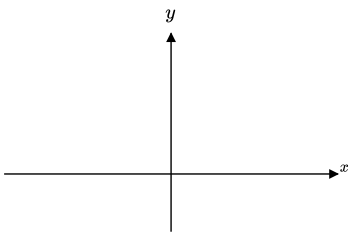
Solve $x^2 + 2x - 15 \leq 0$

$(x + 5)(x - 3) \leq 0$

Step 1: Get 0 on one side

Step 2: Factorise

Step 3: Sketch and reason



Again, what can we say about the x value of any point in this region?

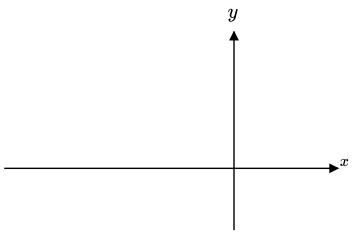
1.20 Inequalities

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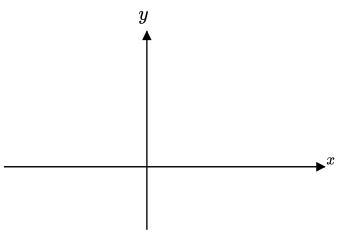
Further Examples

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Solve $x^2 + 5x \geq -4$



Solve $x^2 < 9$



Note: do not skip the 'sketch step'.

To use the quadratic inequality solver on ClassWiz, go to
Menu → Inequalities, then choose 'order 2' (i.e. quadratic)
Try the same on a CG50.

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Test Your Understanding

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Edexcel C1 May 2010 Q3

Find the set of values of x for which
(a) $3(x-2) < 8 - 2x$, (2)
(b) $(2x-7)(1+x) < 0$, (3)
(c) both $3(x-2) < 8 - 2x$ and $(2x-7)(1+x) < 0$. (1)

(a)

(b)

(c)

Edexcel C1 June 2008 Q8

Given that the equation $2qx^2 + qx - 1 = 0$, where q is a constant, has no real roots,
(a) show that $q^2 + 8q < 0$. (2)
(b) Hence find the set of possible values of q. (3)

(a)

(b)

1.20 Inequalities

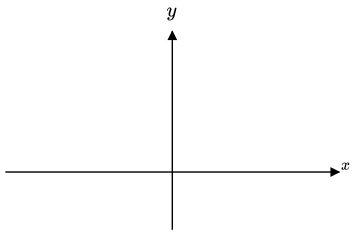
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Deal with inequalities with a division by x

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Find the set of values for which $\frac{6}{x} > 2, \quad x \neq 0$

Why can't we just multiply both sides by x?



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Exercise 3D/3E – Y1 Page 47-48, 50-51

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3 Use set notation to describe the set of values of x for which:

g $4(3x+7) \leq 20$ or $2(3x-5) \geq \frac{7-6x}{2}$

6 Find the set of values of x for which $x^2 - 5x - 14 > 0$.

1.20 Inequalities

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Exercise 3D/3E – Y1 Page 47-48, 50-51

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8 Given that $x \neq 3$, find the set of values for which $\frac{5}{x-3} < 2$.

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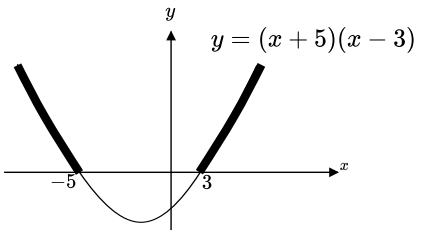
9 The equation $kx^2 - 2kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < 3$.

1.20 Inequalities

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Inequalities on Graphs

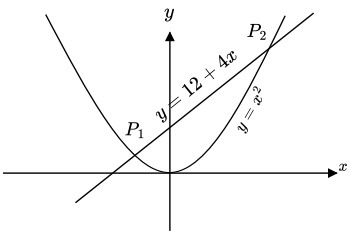
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When we solved quadratic inequalities, e.g. $(x + 5)(x - 3) > 0$
We plotted $y = (x + 5)(x - 3)$ and observed the values of x for which $y > 0$.

Can we use a similar method when we don't have 0 on one side?

Example: L_1 has equation $y = 12 + 4x$. L_2 has equation $y = x^2$.
The diagram shows a sketch of L_1 and L_2 on the same axes.
a) Find the coordinates of P_1 and P_2, the points of intersection.
b) Hence write down the solution to the inequality $12 + 4x > x^2$.



- Solve simultaneously to find points of intersection:
-